

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

Total Marks:25

Comparative Analysis

Code of Conduct:

Abangithan S /

I 192521299 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding ; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Star topology Vs Bus topology

Introduction:

→ Star and bus are two types of network topologies used to connect computers and other devices. They mainly differ in cost, reliability, performance, installation and maintenance.

→ In terms of reliability, star topology is better because failure of one node or its cable does not affect the other nodes. In bus topology, failure of the main backbone cable can bring down the entire network.

→ Star topology also provides better performance because communication is managed through a central switch, which reduces collisions. In bus topology, all devices share the same communication medium, so performance decreases when more devices are added.

Conclusion:

→ Star topology is more reliable and provides better performance, while bus topology is cheaper and simpler to install.

2. Mesh topology vs star topology - Fault tolerance.

→ Mesh topology provides very high fault tolerance because each node has multiple connections to other nodes. Therefore, there are multiple paths available for data transmission.

→ If one link fails in a mesh network, data can travel through an alternate path without interrupting communication.

→ Star topology has moderate fault tolerance. Failure of an individual node does not affect other devices, but failure of the central switch or hub can stop communication for the entire network.

→ Therefore, mesh topology is more robust and reliable than star topology for critical environments where continuous communication is required.

Conclusion:

→ Mesh topology provides higher fault tolerance because it has multiple alternative paths, whereas star topology depends on its central device.

Bus topology vs mesh topology - complexity of installation

→ Bus topology is simple and easy to install because it uses a single backbone cable with a small number of connections, it requires less cabling and networking hardware, making it suitable for small networks.

→ Mesh topology is much more complex to install because each node must be connected to multiple or every other node. This requires a large number of cables and ports.

→ As the number of devices increases the complexity and cost of a mesh network increase significantly. Installation and maintenance also become more time consuming.

Conclusion:

→ Bus topology is simpler, cheaper and easier to install, while mesh topology is more complex, expensive and difficult to implement.

4 Hybrid topology vs star topology - scalability

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→ Hybrid topology is highly scalable because it combines two or more different topologies, such as star, mesh and bus. Different sections of a network can use different topologies according to their requirements.

→ New network segments can be added without significantly affecting the existing network structure.

→ star topology is also scalable, but its expansion depends on the available ports of the central switch or hub. When all ports are occupied, additional switches or hardware are required for further expansion.

Conclusion :

→ Hybrid topology provides scalability and flexibility than star topology because different network structure can be combined and expanded according to organizational requirements.

5. star, mesh, bus and Hybrid topology - maintenance

→ Star topology is relatively easy to maintain because the central switch makes it easier to identify and isolate problems. If a particular cable or node fails, the fault can usually be located without affecting the entire network.

→ Mesh topology is more difficult to maintain because it contains a large number of connections. Troubleshooting and managing these connections can be time consuming.

→ Bus topology is also difficult to maintain because a fault in the main backbone cable can affect the entire network, and locating the exact fault can be difficult.

→ Hybrid topology has moderate maintenance complexity. It provides flexibility but combines multiple topologies so, skilled administrators and careful monitoring are required.

conclusion :

→ star topology is the easiest to maintain, while mesh topology is the most complex. Bus topology has difficulty in locating backbone faults, and hybrid topology requires skilled administration because it combines different network structure.

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